

ADVANCE Network-Diffusion Study

Peer exposure, threshold, susceptibility and infectiousness in adolescent e-cigarette adoption (W1–W10)

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June 15, 2026

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1. Introduction

This is the threshold/diffusion companion to the ADVANCE *disadoption* study. Where the disadoption paper models the **1** $\rightarrow$ **0** transition (quitting), this project models the **0** $\rightarrow$ **1** transition (adoption) on the same USC ADVANCE panel. The lead analyst is Y. Cao; the current pipeline is [codes/260612_netdiffuseR_past6mo_revised13.R](#) on `main`.

Scope (2026-06-15). The study is now focused on the regressions reported in `docs/260612_threshold_report_final.pdf`, namely:

1. **Adoption** — a mixed-effects discrete-time hazard (random intercepts for student and school), with a school-fixed-effects logistic regression as a sensitivity check (report Tables 2–3).
2. **Threshold** — the degree-weighted grouped-binomial model (report Tables 2–3).
3. **Peer-independent subgroups** — logistic “membership” models for users (No Exposure, No Perceived Friends, Neither, Low Threshold) and non-users (High Exposure, High Perceived Friends, Both High), including a social-media exposure term (report Tables 1a–1b).

The susceptibility and infectiousness regressions of the earlier 260603 pipeline have been **retired** from the current report; their specifications are kept in §3 for the record but no longer carry results. The substantive question (per T. Valente, 2026-06-03) is the **peer-independent adopter** — students who adopt with no network exposure, no perceived friends, or a low threshold — and whether **social-media exposure** explains them.

All regression cells report **odds ratios** $\exp(\hat{\beta})$ with the model p-value. Numbers in this document were reproduced end-to-end on the 042326 release with **netdiffuseR 1.25.0** (see §6). Two corrected, AO-labelled re-analyses accompany the report ([codes/07-threshold-corrected-A0.R](#), [codes/08-socialmedia-power-A0.R](#)); their results are in §5.

2. Data

2.1 Sources and sample

- **W1–W8:** `ADVANCE_W1-W8_Data_Complete_042326.xlsx` — 4,437 students $\times$ 10,341 columns.
- **W9–W10 HS:** `ADVANCE_W9-W10_HS_Data_Complete_042326.xlsx` — 1,060 students (subset of W1–W8).
- These are the **same physical files** the disadoption study uses (`A - Network-Disadoption/data/advance/Data/`). The new 260603 pipeline reads the consolidated XLSX directly (slicing per-wave columns `w{w}_*`); the older 260427 pipeline read per-wave `WN_adv_data.csv`. The friendship edges are parsed in-script from the `w{w}_friend{k}_school` nomination columns.

2.2 Cohorts, schools and the grade-period timeline

Fourteen schools in two cohorts. School 108 is excluded; schools with < 20 students are dropped. Ethnic composition (`sch_type`) is assigned by school: Asian-majority {103,105,112,113,212,213}, Hispanic-majority {102,106,107,114,214}, Other otherwise.

Cohort	Schools	Waves used	Grade-period (GP) map
C1 early	101–105	W3–W8	GP = W
C1 late	106,107,112,113,114	W3–W8	GP = W
C2	201,212,213,214	W5–W10	GP = W – 2

Both cohorts are aligned on **six grade-periods**, $GP3\text{--}GP8 = \{10\text{th Fall}, 10\text{th Spring}, 11\text{th Fall}, 11\text{th Spring}, 12\text{th Fall}, 12\text{th Spring}\}$. Grade 9 (GP1–GP2) is dropped so that C1-late and C2 schools, which only enter at 10th-Fall, are comparable. Combined panel: **4,007 students, 841 adopters (21.0% by 12th-Spring)**.

2.3 Diffusion network construction

For each school s a directed friendship network is built per grade-period from the nomination columns; ties are kept only between consenting panel members. The per-school diffnet is assembled with `as_diffnet(graph, toa, ...)` and the 14 are merged with `do.call(c, ...)` into the combined object G on which all network statistics are computed. Past-6-month use is **monotonised** (once 1, stays 1), and the time-of-adoption is $\tau_i = \min\{t : y_{it} = 1\}$.

2.4 Covariates

Demographics are taken at each student’s **baseline wave** (first wave their school appears): `female = 1[dem_gender = 0]`, `hispanic = 1[eth = 1]`, `asian = 1[race = 2]`, `sex_min = 1[dem_sexuality ≠ 1]`. Time-varying covariates merged at the relevant grade-period: `gad = rcads_gad_mean`, `mdd = rcads_mdd_mean`, `friends_ecig = friends_use_ecig` (perceived friend use, “6” recoded NA). Social-media exposure (`sm_post_1..6`, `ecig_posted_1..3`) exists **only W4–W8** — a fact that dominates §6.

2.4.1 Parent education — harmonisation, and how it differs from the disadoption study

Parent education needs cross-wave handling because the questionnaire scale changed mid-panel. Per the codebook (042326):

- **Legacy** `w{w}_dem_high_par_edu`: 1=≤8th grade, 2=some HS, 3=HS grad, 4=some college, 5=college grad, 6=advanced degree, **9=Don’t know**.
- **New** `w{w}_dem_high_par_edu_new` (W7–W10 only): 1..4 as above, 5=vocational, 6=associate, 7=bachelor’s, 8=master’s, 9=Don’t know.

The 260603 pipeline reads the **legacy-named** column at all waves, remaps W7+ with `c(1,2,3,4,4,4,5,6,NA)`, and then assigns each student their **first valid value across all waves** (a single, effectively time-invariant number).

This is *not* equivalent to the disadoption construction, in three ways:

	Disadoption (R/01-advance-panel.R)	Diffusion (260603)
Fill rule	LOCF per student-wave (time-varying)	first-valid across waves (one static value)
“Don’t know” (9) at W7+	9 → 7 (kept as top level)	9 → NA (dropped)
“Don’t know” (9) at W1–W6	kept as 9	kept as 9

Two consequences worth flagging:

1. **A latent scale bug, shared by both.** Empirically, in the 042326 release the legacy-named column `w{w}_dem_high_par_edu` is on the **legacy 6-level scale at every wave** (values $\{1, \dots, 6, 9\}$; no 7s or 8s, even at W7–W8 — the new scale lives in the tiny `..._new` column, $n = 98$ at W7). So the W7+ “new→legacy” remap is applied to data that is *already* legacy-scaled: its 5→4, 6→4 rules **mis-collapse “college graduate” and “advanced degree” into “some college”** for any value drawn from W7–W10. Impact is bounded by which wave supplies each student’s value (most come from an early legacy wave), but it is a real defect in both pipelines.
2. **“Don’t know” leaks as a numeric level in this pipeline.** Because $9 \rightarrow \text{NA}$ only fires for $w \geq 7$, and *first-valid* almost always pulls from an early ($w < 7$) wave, the value **9 survives** and enters the GLMs as a continuous predictor ranked **above** “advanced degree” (6). In the threshold sample, **60 of 835 adopters (7.2%) have `par_edu` = 9**. This contaminates any model that treats `par_edu` as continuous — and `par_edu` is “significant” in exactly the two models where it should be least expected (susceptibility OR 1.18, $p = .012$; infectiousness OR 1.25, $p = .016$), which is a red flag rather than a finding.

Recommended fix: recode $9 \rightarrow \text{NA}$ at *all* waves; since the read column is legacy-scaled throughout this release, drop the W7+ new-scale remap (or read `..._new` only where populated); and treat `par_edu` as an ordered factor (or 1–6 continuous with DK = NA).

2.5 Temporal alignment — are the regressions lagged?

Short answer: **only the adoption model is lagged**. The four model families differ:

Model	Predictor timing	Outcome timing	Lagged?
Adoption / exposure	period t	adoption at $t + 1$	Yes (one period)
Threshold	at TOA τ_i	at TOA τ_i	No (contemporaneous)
Susceptibility	at TOA τ_i	at TOA τ_i	No (contemporaneous)
Infectiousness	at TOA τ_i	at TOA τ_i	No (contemporaneous)

The adoption model is a discrete-time hazard: exposure and covariates measured at t predict *next-period* adoption (`adopt_next = adopt_mat[, t+1]`), so the predictor strictly precedes the event. The threshold/susceptibility/infectiousness models put **one row per adopter at their adoption period** and merge `gad`, `mdd`, `friends_ecig`, demographics at τ_i **itself**. The *network* quantity is inherently ordered (threshold = friends already using *before* TOA; susceptibility = friends who adopted in $\tau_i - 1$; infectiousness = friends adopting in $\tau_i + 1$), but the **explanatory covariates are concurrent with adoption**, which raises a simultaneity caveat: e.g. mental-health state *at* the moment of adoption may be a consequence, not a cause. In the disadoption study predictors are taken at $w - 1$; matching that convention here (covariates at $\tau_i - 1$) would make the two studies consistent and reduce the reverse-causation risk.

3. Methods — model specifications

Let $G = (V, E)$ be the combined directed diffnet, $t \in \{3, \dots, 8\}$ the grade-periods, $y_{it} \in \{0, 1\}$ monotone past-6-month use, $\tau_i = \min\{t : y_{it} = 1\}$ the time of adoption, $A_i^{\text{out}}(t) = \{j : (i \rightarrow j) \in E_t\}$ ego i 's out-neighbours, and $E_{it} = \frac{\sum_{j \in A_i^{\text{out}}(t)} y_{jt}}{|A_i^{\text{out}}(t)|}$ the network exposure (share of alters already adopted). $\mathbf{x}_i$ is the covariate vector (cohort, female, hispanic, asian, par_edu, gad, mdd, friends_ecig, sex_min), α_t a grade-period fixed effect, and ξ_s a school fixed effect (C2 schools collapsed to the reference so `factor(school_c1fe)` yields only C1 dummies, with cohort absorbing the C1-vs-C2 mean).

3.1 Threshold — grouped binomial

For each non-isolate adopter ($n_i = |A_i^{\text{out}}(\tau_i)| > 0$), let $k_i = |\{j \in A_i^{\text{out}}(\tau_i) : \tau_j < \tau_i\}|$ be the number of out-alters who adopted **strictly before** i . The empirical threshold is $\hat{T}_i = k_i/n_i$. We model the counts directly:

$$k_i | n_i \sim \text{Binomial}(n_i, \pi_i), \quad \text{logit}(\pi_i) = \alpha_{\tau_i} + \xi_{s(i)} + \mathbf{x}_i^\top \beta.$$

In R: `glm(cbind(k_users, n_alters - k_users) ~ factor(toa) + cohort + female + hispanic + asian + par_edu + factor(school_c1fe) + gad + mdd + friends_ecig + sex_min, family = binomial)`. Isolates are excluded because $\hat{T}_i = 0/0$ is undefined. This **degree-weighted** binomial is the re-specification adopted from the disadoption collaborator's 2026-05-12 diagnostic: it replaces a quasibinomial on the raw proportion $\hat{T}_i$, which over-weights low-degree adopters whose threshold lives on a coarse rational grid $\{0, \frac{1}{2}, \frac{1}{3}, \dots\}$.

3.2 Adoption — discrete-time exposure hazard

Person-period rows (i, t) for every still-at-risk student ($y_{it} = 0$); predictors at t , outcome at $t + 1$. The **current report's main model (Table 2)** is a *mixed-effects* logistic hazard with random intercepts for student and school:

$$\Pr(y_{i,t+1} = 1 | y_{it} = 0) = \text{logit}^{-1}(\gamma E_{it} + \alpha_t + \mathbf{x}_i^\top \beta + u_i + v_{s(i)}), \quad u_i \sim \mathcal{N}(0, \sigma_u^2), \quad v_s \sim \mathcal{N}(0, \sigma_v^2).$$

`lme4::glmer(adopt_next ~ exposure + factor(grade_period) + ... + gad + mdd + friends_ecig + sex_min + (1 | id) + (1 | school), family = binomial)`. The **sensitivity model (Table 3)** replaces the random effects with school fixed effects so individual school ORs are inspectable:

$$\Pr(y_{i,t+1} = 1 | y_{it} = 0) = \text{logit}^{-1}(\gamma E_{it} + \alpha_t + \xi_{s(i)} + \mathbf{x}_i^\top \beta).$$

The mixed model returns *subject-specific* ORs (larger in magnitude, e.g. exposure 6.46) while the fixed-effects GLM returns *population-average* ORs (exposure 2.31); both agree on sign and significance for every predictor.

3.3 Susceptibility and infectiousness — quasibinomial (*retired from current scope*)

These were computed in the earlier 260603 pipeline but are **not part of the current report**; the specification is kept here for the record only. One row per adopter; **susceptibility** is the share of i 's out-alters who adopted *immediately before* i (excludes $\tau_i = 3$), **infectiousness** the share adopting *immediately after* (excludes $\tau_i = 8$), each a proportion in $[0, 1]$ modelled with an over-dispersed logit $\text{logit } \mathbb{E}[S_i] = \alpha_{\tau_i} + \xi_{s(i)} + \mathbf{x}_i^\top \beta$, $\text{Var}(S_i) = \phi \mu_i(1 - \mu_i)$ (family = quasibinomial).

3.4 Subgroup membership — “Tom’s table”

Seven subgroups — four among **users** (report Table 1a), three among **non-users** (Table 1b) — each modelled as binary membership within its stratum:

$$\text{logit } \Pr(m_{ic} = 1) = \alpha_{\tau_i} + \mathbf{z}_i^\top \theta_c,$$

where $\mathbf{z}_i$ stacks demographics, GP3/GP8 degree, perceived friend use, exposure, and a social-media term; α_{τ_i} are grade-period fixed effects. The predictor that *defines* a column is excluded from that column’s model (definitional overlap). **Users:** *No Exposure* $E_{i\tau_i} = 0$; *No Perceived Friends* $\text{PFU}_{\tau_i} = 0$; *Neither* both = 0; *Low Threshold* $\hat{T}_i \leq 0.25$ (the sample’s 70th percentile, per Valente 1996). **Non-users:** *High Exposure* $\max_t E_{it} > 0$; *High Perceived Friends* $\max_t \text{PFU}_{it} > 0$; *Both High* both.

3.5 The social-media term

In the current report the social-media exposure indicator SM_i (`sm_ecig_any` at TOA = saw e-cig content on any of six platforms, W4–W8) enters the **user subgroup model** of §3.4 as one more covariate, and its OR $\exp(\hat{\eta})$ is the `sm_ecig_any` row of Table 1a:

$$\text{logit } \Pr(m_{ic} = 1) = \eta \text{SM}_i + \alpha_{\tau_i} + \mathbf{z}_i^\top \theta.$$

(An older standalone 18-model sensitivity sweep — six SM operationalisations $\times$ three subgroups — also exists in the pipeline but is not the reported result.)

4. Results — what we obtained re-running the pipeline

These are **our** numbers from re-running `codes/260612_netdiffuseR_past6mo_revised13.R` on the 042326 release (netdiffuseR 1.25.0), laid out in the **same tables** as `260612_threshold_report_final.pdf`. Each table is **not one regression but several** — we state the model behind each column, then interpret it in plain words. $** = p < .01$, $* = p < .05$.

4.1 Tables 2–3 — Adoption and Threshold side by side

Each of Tables 2 and 3 juxtaposes **two different regressions**: an *adoption* model (does a still- non-using student start next period?) and the *threshold* model (among those who start, how built-up was the peer pressure?). Table 2 uses the **mixed-effects** adoption model, Table 3 the **school-fixed-effects** version; the threshold column is the same grouped binomial in both.

The regressions (math). Adoption is a discrete-time hazard over person-periods (i, t) with E_{it} = share of i 's friends already using:

$$\textbf{Table 2 (mixed): } \text{logit } \Pr(y_{i,t+1}=1 \mid y_{it}=0) = \gamma E_{it} + \alpha_t + \mathbf{x}_i^\top \beta + u_i + v_{s(i)},$$

$$\textbf{Table 3 (FE): } \text{logit } \Pr(y_{i,t+1}=1 \mid y_{it}=0) = \gamma E_{it} + \alpha_t + \xi_{s(i)} + \mathbf{x}_i^\top \beta.$$

Threshold is the degree-weighted grouped binomial over adopters (n_i friends, k_i already using):

$$\textbf{Threshold: } k_i \mid n_i \sim \text{Binomial}(n_i, \pi_i), \quad \text{logit } \pi_i = \alpha_{\tau_i} + \xi_{s(i)} + \mathbf{x}_i^\top \beta.$$

Our numbers (Adoption ORs; Threshold ORs, n = 406 non-isolate adopters):

Predictor	Adoption — Table 2 mixed	Adoption — Table 3 FE	Threshold (Tables 2–3)
Network exposure	7.6 **	2.31 **	—
exp(γ)			
Perceived friend use	1.72 **	1.43 **	1.20 **
MDD	(unstable)	1.61 **	0.81
GAD	(unstable)	0.77 *	0.97
Cohort C2	(unstable)	1.10	2.64 *
Asian	(unstable)	0.77	0.60 (.053)
Hispanic / female / par_edu / sex_min	(unstable)	ns	ns
School 104 / 114 (Hispanic-maj.)	—	ns	2.99* / 3.63**
Constant; GP4 ... GP8	GP8 separates	GP8 separates	−20.6; 17.2 ... 18.8 (degenerate)

Two reproduction notes. (1) The **FE adoption model** (Table 3) reproduces the report exactly — exposure 2.31, MDD 1.61, GAD 0.77, PFU 1.43. (2) The **mixed model** (Table 2) is numerically **unstable**: on identical data our `glmer` converged to a different point (the demographic ORs blow up and GP8 separates), so we trust Table 3 and treat the mixed ORs as not robustly identified — a first sign that the sparse, separated design strains these models. The Threshold column’s **Constant and all GP rows are a separation artifact** (the $-20.6 / 17.2 \dots 18.8$ block in the report) — see §5.1.

For grandma. *Whether a teen starts vaping in the next half-year depends most on how many of their friends already vape — a kid surrounded by vapers is about **2.3**× **more likely** to start. Feeling depressed nudges them toward it; feeling anxious, slightly away from it; thinking your friends vape adds a push too. Now, among the kids who do start, the “threshold” asks how many friends were already vaping at that moment: kids who start in 12th grade waited until far more friends were already vaping, while early-starters jumped in with almost no one — and believing your friends vape goes with waiting for a higher bar. (The giant numbers in the grade rows are a computer hiccup, not a real result; we fix it in §5.)*

4.2 Tables 1a–1b — Peer-independent subgroups (each column is its own regression)

Every column of Tables 1a/1b is a **separate logistic regression**: among users (1a) or non-users (1b), it asks which traits predict belonging to that “peer-independent” profile, controlling for grade-period:

$$\text{logit Pr}(\text{member of subgroup } c) = \alpha_{\tau_i} + \mathbf{z}_i^\top \theta_c.$$

The **social-media term** `sm_ecig_any` sits inside $\mathbf{z}$ for the user columns; it is undefined for non-users (no time-of-adoption) and shown as “—”.

Table 1a — users (our ORs; $p < .01$, * $p < .05$):**

predictor	No Exposure	No Perc. Friends	Neither	Low Threshold
Asian	2.49 (.05)	2.31	3.92 *	3.53 **
MDD	1.07	0.53	0.46 *	0.79
Out-degree (GP3)	0.81 *	1.10	0.94	0.98
Perceived friends (GP3)	0.74 *	0.81	0.51 **	0.86

predictor	No Exposure	No Perc. Friends	Neither	Low Threshold
Perceived friends (GP8)	0.85	0.56 **	0.57 **	0.85
sm_ecig_any (W4–W8)	2.31 *	0.40 *	0.87	1.39

Table 1b — non-users / “resisters” (our ORs):

predictor	High Exposure	High Perc. Friends	Both High
Asian	0.51 **	0.34 **	0.37 **
Female	1.03	1.30 **	1.63 **
MDD	1.20	1.51 **	1.53 **
Out-degree (GP3)	1.39 **	1.06 *	1.25 **
Perceived friends (GP8)	1.47 **	—	3.42 **
par_edu	0.96	0.93 **	0.98

These reproduce the report exactly. Note **par_edu** looks significant in the “High Perceived Friends” column (0.93**) — a red flag we trace to a coding bug and fix in §5.3.

For grandma. *We sorted the teens into “types”. Among vapers who had **no vaping friends** in the network, Asian kids are over-represented and — surprisingly — those who’d seen vaping on social media are **2.3× more likely** to be in this no-friends group (as if social media stood in for friends). Among teens who **don’t** vape even though vaping surrounds them (the “resisters”), Asian kids resist strongly, while girls and kids who feel depressed resist **less**. One number about parents’ education looks meaningful, but it turns out to be an artifact of how “Don’t know” answers were coded — we clean that up next.*

5. Two corrected re-analyses (_A0)

The two issues below are reproduced from the current report and then **corrected**, with the fix shipped as a labelled script. Each is explained intuitively first, then technically with the relevant algebra.

5.1 Threshold — the grade-period block is spurious (complete separation)

Correction: `codes/07-threshold-corrected-A0.R` → `outputs_A0/model/threshold_corrected-A0.csv`

Intuitively. A “threshold” answers: *of the friends you could be influenced by, how many were already using when you started?* The model needs, for each adopter, a count of friends-already-using k_i out of friends-total n_i . But the network only starts at grade-period 3 (GP3). For someone who adopts *at* GP3 there is no earlier period in the data, so **none** of their friends can have adopted before them: $k_i = 0$ for **every** GP3 adopter (149 non-isolates; 44 survive into the complete-case fit — all with $k = 0$ either way). When an entire time category contains only zeros, the model can fit it perfectly by sending that category’s coefficient to $\pm\infty$. That is exactly what happens: the report’s threshold column shows a Constant of **−20.63** and grade-period rows of **17.23, 17.88, 18.11, 18.46, 18.84** — these are not findings, they are the optimiser running off to infinity (ORs of 10^7 – 10^8). Yuchan already spotted the *same* pathology for School 105 and dropped it; the grade-period block has it too, undiagnosed — **and a second copy survives in School 105 even after GP3 is removed.**

Technically. The model is the grouped binomial of §3.1, $k_i \mid n_i \sim \text{Binomial}(n_i, \pi_i)$, $\text{logit } \pi_i = \alpha_{\tau_i} + \xi_{s(i)} + \mathbf{x}_i^\top \beta$, with GP3 the reference level absorbed into the intercept. Because $\tau_i = 3 \Rightarrow k_i = 0$, the reference stratum is *perfectly separated*: the likelihood increases monotonically as $\alpha_{\text{GP3}} \rightarrow -\infty$ with the $\text{GP} \geq 4$ dummies diverging to compensate; the MLE does not exist (printed SE ≈ 750). The same holds for School 105 (9 rows, all $k = 0$). Two fixes:

- **B. Exclude GP3 adopters** ($\tau_i > 3$), as the retired susceptibility model already did. This is information-lossless for the slopes — algebraically, $B_{\text{intercept}} - A_{\text{intercept}}$ equals A’s GP4 coefficient and B’s GP dummies are A’s GP *differences* — so every substantive OR is bit-identical to the as-is fit (max difference 10^{-14}). *But it leaves the School-105 separation in place.*
- **C. Firth-penalized GLM** (brglm2, guaranteed-finite estimates) on the **full** 406 adopters — **the recommended primary fix**: it keeps all the data, resolves *both* separations, and recovers the genuine steep grade-period trend. Its bias-reducing shrinkage moves a few coefficients at the 2nd–3rd decimal, so the ORs are *substantively similar*, not bit-identical.

parameter	A. as-is (n=406)	B. excl. GP3 (n=362)	C. Firth, full (n=406)
GP4 / GP5 / GP8	10^7 – 10^8 (<i>sep.</i>)	— / 1.91* / 5.02**	40.7** / 76.4** / 196.7**
Perceived friend use	1.20 ***	1.20 ***	1.20 ***
Asian	0.598 (.05)	0.598 (.05)	0.604 (.05)
Cohort C2	2.64 *	2.64 *	2.43 *
Hispanic / GAD / MDD / sex_min	ns	ns	ns
separation remaining	Constant + all GP + Sch105	Sch105	none

The substantive ORs agree across all three (PFU ≈ 1.20 , Asian ≈ 0.60 , Cohort ≈ 2.5) — the separation never threatened those conclusions — but only C produces a fully interpretable, separation-free fit. The corrected story is itself substantive: **the adoption threshold rises steeply across grade-periods** (later adopters require far more friends already using), which the degenerate as-is block could not express. Tables 2–3 should report **model C**; the -20.63 / $17.x$ rows must be removed.

5.2 Social media — fragile and spec-dependent, on an under-powered design

Correction: `codes/08-socialmedia-power-A0.R` $\rightarrow$ `outputs_A0/model/socialmedia_corrected-A0.csv`

Intuitively. The question is whether *social-media* e-cig exposure explains adoption among peer-independent students. The catch: the social-media items exist only in **W4–W8**, so the 224 students who adopt at GP3 (= W3) — the single largest cohort — have **no** measurement at adoption and drop out (only **456 of 841 adopters, 54%**, are measurable). Table 1a then shows two significant cells — No Exposure **OR 2.31** and No Perceived Friends **OR 0.40** — pointing in *opposite* directions, which already warns that this is not one clean “alternative channel”. But the right reading is **not** “artifact”: the No-Exposure association is **fragile and specification-dependent**. It survives a Firth penalty (OR 2.12, $p = .021$, so it is not an over-fitting blow-up), yet it attenuates to **borderline** (OR ≈ 1.5 , $p \approx .10$) once we adjust for *temporally-valid baseline confounders* instead of the model’s `friends_ecig_toa` term — which is measured **at the moment of adoption**, i.e. post-treatment relative to the outcome, so controlling for it risks collider bias. Whether the No-Exposure effect is “real” depends on a confounder-vs-collider judgement the data cannot settle. The No-Perceived-Friends cell, by contrast, is robust *and* protective *and* opposite-signed — which is the opposite of what the channel hypothesis predicts.

Technically. With $\mathcal{A}$ the adopters and $\mathcal{A}_{\text{SM}} = \{i : \text{SM}_i \text{ observed at } \tau_i\}$, collection only in $t \in \{4, \dots, 8\}$ gives $\tau_i = 3 \Rightarrow \text{SM}_i$ missing and $|\mathcal{A}_{\text{SM}}|/|\mathcal{A}| = 456/841 = 0.54$, concentrated in the earliest cohort (non-ignorable). The “collapse” of the No-Exposure cell decomposes into (a) covariate adjustment and (b) a $239 \rightarrow 455$ listwise-deletion sample change — on a *fixed* sample the parsimonious OR is 1.52, not 1.26 — so it is not a single clean effect. The load-bearing covariate is `friends_ecig_toa` (post-treatment); using only baseline GP3 confounders gives the fair middle estimate. As a power floor, the post-hoc Wald minimum detectable OR, $\text{OR}_{\min} = \exp[(z_{.975} + z_{.80}) \widehat{\text{SE}}]$, is $\approx$ **2.0–2.2** for the null cells: those nulls are *uninformative*, not evidence of absence. Specification ladder (each cell OR (p); EPV = events-per-variable):

subgroup	1. full (Yuchan)	2. Firth	3. baseline GP3 (<i>fair</i>)	4. parsimonious	p_{Bonf}
No Exposure	2.31 (.013), EPV 7	2.12 (.021)	1.51 (.10) , EPV 19	1.26 (.25)	.050
No Perceived Friends	0.40 (.010)	0.43 (.017)	0.55 (.017)	0.51 (.001)	.041
Neither	0.87 (.70)	0.87 (.68)	0.89 (.66)	0.67 (.08)	1.00
Low Threshold	1.39 (.31)	1.35 (.35)	1.11 (.67)	0.95 (.82)	1.00

(A cumulative “ever exposed W4..TOA” measure recovers 592/841 adopters but is mechanically tied to adoption timing — more waves of opportunity for later adopters — so it is a *different* exposure definition, not a cleaner version of the snapshot; it is reported in the CSV for completeness.) **Conclusion.** The only social-media association stable across all specifications is the **protective, opposite-signed** No-Perceived-Friends cell, which does *not* support social-media-as-alternative-channel; the headline No-Exposure cell is **borderline and specification-dependent**; and the genuinely peer-independent cells stay null with a detectable-OR floor near 2. The honest statement is “fragile and under-powered — not yet answerable”, not “social media has (or lacks) an effect”. A clean test needs a pre-registered baseline-confounder block, a pooled W4–W8 exposure measure, and the GP3 adopters acknowledged as structurally unmeasurable.

(*Aside: the Hispanic threshold coefficient is null under school FE because individual ethnicity is collinear with the Hispanic-majority school dummies — the signal sits in Schools 104/114. Worth keeping in mind when reading Tables 2–3; not pursued in detail here.*)

5.3 The report’s tables, with every fix applied

Correction: [codes/09-fully-corrected-tables-A0.R](#) → [outputs_A0/model/fully_corrected_*-A0.csv](#)

This section re-runs the report’s four tables with **all** the fixes at once: the **par_edu** recode below, plus **Firth** for every model that suffered separation or small-sample fragility (threshold, adoption, and the subgroup/social-media logistics).

The par_edu fix. Per the codebook the parent-education column is the **legacy 6-level scale at every wave** (1 = ≤8th grade ... 6 = advanced degree, **9** = “Don’t know”). Yuchan’s pipeline (a) left “Don’t know” in as a numeric **9** — ranked *above* “advanced degree” — for the 7% of adopters who gave it, and (b) applied a “new-scale” remap at W7+ that does not match this release. The corrected recode homogenises the scale to a clean $\{1, \dots, 6\}$ and sends “Don’t know” to missing:

$$\text{par_edu}_i^{\text{corr}} = \text{first valid } v \in \{1, \dots, 6\} \text{ across the student’s waves,} \quad 9 \mapsto \text{NA.}$$

Corrected Tables 2–3 (Adoption — Firth FE; Threshold — Firth grouped binomial):

Predictor	Adoption (Firth FE)	Threshold (Firth)
Network exposure	2.34 **	—
Perceived friend use	1.44 **	1.20 **
MDD	1.61 **	0.78
GAD	0.79 *	1.02
Cohort C2	1.05	2.34 (.06)
Asian	0.80	0.61 (.06)
par_edu (recoded)	0.98 (<i>ns</i>)	1.04 (<i>ns</i>)
Constant; GP4 / GP5 / GP8	GP8 = 0.007** (<i>finite</i>)	40** / 70** / 189** (<i>finite</i>)

The degenerate blocks are gone: the threshold grade-period effects are now **finite, large and increasing** (40 → 189), i.e. the genuine result the separation hid — *later adopters start only once many more friends*

already use. Every substantive OR is unchanged (exposure 2.34, PFU 1.20–1.44, MDD 1.61, GAD 0.79, Asian ≈ 0.6).

Corrected Tables 1a–1b (subgroups — Firth + recoded par_edu): the conclusions of §4.2 hold — Asian elevated among peer-independent users (Neither **3.45***, Low Threshold **3.25***) and strongly protective among resisters (**0.51 / 0.34 / 0.36**, all **); MDD/female raise resistance. Two changes matter:

	as-is (§4.2)	corrected (§5.3)
par_edu in “High Perceived Friends”	0.93 ** (looked real)	1.01 (ns) — spurious effect removed
sm_ecig_any in “No Exposure”	2.31 *	2.40 ** (survives Firth)

So the par_edu “effect” was manufactured by the “Don’t know” coding bug and **vanishes** once it is recoded. The social-media “No Exposure” cell, by contrast, **survives** Firth — confirming (as §5.2 already showed) that it is *not* a small-sample artifact; its fragility lives in the confounder choice (friends_ecig_toa is measured at adoption), which Firth does not address. Net: Firth fully fixes the threshold; for social media it removes one doubt (small-sample) but leaves the design under-powered and the No-Exposure estimate confounder-dependent.

For grandma. *We re-ran everything with the dials set correctly. First we fixed how “parents’ education” was recorded: people who answered “I don’t know” had been counted as if that were the **highest** education level — so we moved those to “missing”. The moment we did, a result that looked real (something about parents’ education among the resisters) **disappeared** — it was never real, just a coding slip. Second, we used a sturdier statistical method (Firth) that stops the computer from producing those impossible giant numbers. With it, the real pattern shows cleanly: kids who start vaping later in high school only do so once **many** of their friends already vape. None of the genuine findings changed — friends who vape, feeling depressed, and being surrounded by vaping still drive adoption. The one stubborn loose end is social media: the sturdier method confirms the “no-friends vapers saw more vaping online” result is not a fluke of small numbers, but we still can’t be sure it’s real, because half the kids can’t be measured for it and it leans on a questionable control — so we mark it “promising but unproven”, not “true”.*

6. Reproducibility

Reproduced end-to-end on the 042326 release with **netdiffuseR 1.25.0** (CRAN). The in-development 1.26.0 aborts at hazard_rate because c.diffnet drops the new \$status field; root cause and a minimal reproducer are filed for the maintainer (git-ignored playground/).

```
# 1) Yuchan's pipeline (produces her Tables 1-3 and caches the analytic frames)
Rscript codes/260612_netdiffuseR_past6mo_revised13.R      # data_path → the two ADVANCE_W*.xlsx
# 2) The corrections (consume the analytic frames; write to outputs_A0/model/)
Rscript codes/07-threshold-corrected-A0.R                 # threshold separation (§5.1)
Rscript codes/08-socialmedia-power-A0.R                   # social-media power/robustness (§5.2)
Rscript codes/09-fully-corrected-tables-A0.R               # all fixes at once: par_edu + Firth (§5.3)
```

The _A0 scripts read the analytic frames built by Yuchan’s pipeline (set A0_THR_SUB / A0_CACHE to point at them); script 09 additionally recomputes par_edu from the raw XLSX. None alter her variable construction otherwise.

7. Open items

1. **Adopt the corrected threshold** (§5.1/§5.3): report the Firth fit — the $-20.63 / 17.x$ grade-period block is a separation artifact, not a result.

2. **Re-frame the social-media result** (§5.2): the positive No-Exposure cell survives Firth but is confounder-dependent and the design is under-powered. Pool W4–W8 exposure, use baseline-only confounders, and pre-register.
3. ~~Recode par_edu~~ — **done (§5.3)**: “Don’t know” (9) → NA, homogeneous 1–6 scale. Roll the recode into the upstream pipeline so all tables use it by default.
4. **Lag the threshold covariates** to $\tau_i - 1$ to match the disadoption convention and blunt the simultaneity concern (§2.5).
5. Decide the **Hispanic estimand** (school-FE contextual vs no-FE individual) and state it once.

Companion document: `A - Network-Disadoption/docs/disadoption-study.md`. *Source report:*
`docs/260612_threshold_report_final.pdf`. *Corrected analyses:* `codes/0{7,8,9}-*-A0.R`, `outputs_A0/model/*-A0.csv`.